

TABE Level E Math — Formula Cheat Sheet

Foundational math · whole numbers to basic geometry · certpathpublishing.store

1 2 3 NUMBER SENSE

Place Value & Rounding

- Places: ones, tens, hundreds, thousands...
- **Rounding:** look at the digit to the right — **5 or more rounds up**, less than 5 stays.
- **Order of operations (PEMDAS):** $\times/\div$ before $+/-$, left to right.

Operations & Estimation

- Line up place values to add and subtract.
- Carry when a column is > 9 ; borrow when the top digit is smaller.
- **Estimate** by rounding first to check your answer.

FRACTIONS, DECIMALS, PERCENT

Fractions

- **Add / subtract:** need a common denominator first.
- **Multiply:** $a/b \times c/d = (a \times c)/(b \times d)$
- **Divide:** flip the second, then multiply ($a/b \div c/d = a/b \times d/c$).
- **Simplify:** divide top and bottom by the same number.

Decimals & Percent

- **Percent $\rightarrow$ decimal:** divide by 100 ($25\% = 0.25$).
- **Decimal $\rightarrow$ percent:** multiply by 100.
- **Percent of a number:** percent (as decimal) $\times$ number.
- **Fraction $\rightarrow$ decimal:** divide top by bottom.

MEASUREMENT & GEOMETRY

Perimeter, Area, Volume

- **Perimeter:** add all sides · square = $4 \times$ side
- **Rectangle area:** $L \times W$
- **Triangle area:** $\frac{1}{2} \times \text{base} \times \text{height}$
- **Circle:** $A = \pi r^2$ · $C = 2\pi r$ ($\pi \approx 3.14$)
- **Box volume:** $L \times W \times H$

US Customary Units

- 1 ft = 12 in · 1 yd = 3 ft · 1 mi = 5,280 ft
- 1 lb = 16 oz · 1 ton = 2,000 lb
- 1 gal = 4 qt = 8 pt = 16 cups
- 1 hr = 60 min · 1 min = 60 sec

RATIOS & DATA

Ratios & Rates

- **Ratio** compares amounts: 3 to 7, 3:7, or $3/7$.
- **Proportion:** $a/b = c/d \rightarrow$ cross-multiply.
- **Unit rate:** divide to get "per 1" (miles $\div$ hours = mph).

Averages & Graphs

- **Mean (average):** sum $\div$ how many.
- **Median:** middle number when in order.
- **Mode:** the value that appears most.
- Always read the title, labels, and key on a chart.

 **Test Day:** TABE Level E is foundational — most questions are 1–2 steps. **Estimate first** to catch slips, line up your place values, and check whether the question asks for **perimeter vs. area**. Use scratch paper for every calculation.